EMORY
HEALTHCARE

June 16, 2025

The Honorable Dr. Mehmet Oz
Administrator
Centers for Medicare and Medicaid Services
U.S. Department of Health and Human Services
Attention: CMS-0042-NC
7500 Security Boulevard
Baltimore, MD 21244-1850

Delivered Electronically via: *http://www.regulations.gov*

Re: Response to CMS-ASTP Request for Information on the Health IT Ecosystem (CMS-0042-NC)

Dear Administrator Oz:

Emory Healthcare, part of Emory University, is an integrated academic health care system committed to providing the best care for our patients; educating health professionals and leaders for the future; pursuing discovery in all of its forms, including basic, clinical and population-based research; and serving our community. As the clinical enterprise of the Robert W. Woodruff Health Sciences Center of Emory University, Emory Healthcare remains the most comprehensive health care system in Georgia, providing extensive inpatient and outpatient services to a diverse patient population.

Emory Healthcare appreciates the opportunity to respond to the Request for Information (RFI) on the Health Information Technology (IT) Ecosystem. We commend the Centers for Medicare and Medicaid Services (CMS) and the Assistant Secretary for Technology Policy/Office of the National Coordinator for Health IT (ASTP/ONC) for their continued leadership in advancing interoperability, patient access to care and their data concerning that care, and innovation in digital health. Our response draws from our experience implementing health IT solutions across clinical, research, and operational domains, including our utilization of certified electronic health records technologies, Health Level Seven® (HL7®) Fast Healthcare Interoperability Resources® (FHIR®) and Bulk FHIR, our participation in the Trusted Exchange Framework and Common Agreement (TEFCA) and our journey to a Unified Data Platform (UDP).

We encourage CMS and ASTP/ONC to consider in its review of comments the immense value of directly observing the workflows that affect and are affected by regulations and innovations in the health IT space. We have found that observing operations in real time often leads to new discoveries and insights that can further inform a more efficient, effective and user-friendly future.

With our shared goals in mind, we offer the following summary of key insights and recommendations, followed by detailed responses to selected RFI questions. If you have any questions regarding these comments, please contact our Assistant Vice President for Federal

Affairs, Jessica Davis, at jessica.ann.davis@emory.edu. We look forward to continuing to participate in a health IT landscape that supports the best care for patients.

Sincerely,

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Chief Quality Officer
Emory Healthcare

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VP and Chief Health Informatics Officer, Emory Healthcare
Associate Professor, Internal Medicine, Emory School of Medicine

EXECUTIVE SUMMARY

Data Quality and Interoperability: We strongly support CMS's emphasis on improving data quality and interoperability. Health care is always seeking ways to better surface the right data at the right time. This improves outcomes, reduces duplicate testing, and supports continuity of care. Improvements in technology support data exchanges without sacrificing data quality but are not yet mature enough to support a reduced reliance upon data standards. The current health data ecosystem requires definitions of shared ontologies, mapping or implementation guides, and clear frameworks for determining data quality. Responses below include our experience with the practical application of such guidance, including our journey to a Unified Data Platform (UDP) and the prerequisite data cleaning, mapping and standardization.

Real-Time and AI-Ready Data: It is in all our best interest to avoid using poor quality data as we collectively move to support artificial intelligence/machine learning (AI/ML)-enabled decision support and research. We recommend CMS partner with other federal agencies, like ASTP/ONC and the Food and Drug Administration (FDA), to invest in recommendations on data quality frameworks, tools and – eventually – certification programs that help the health IT ecosystem avoid a "garbage in, garbage out" scenario. The responses below include our experience with assessing and improving the quality of large volumes of data from external data sources, including other providers, medical devices, and health IT systems not certified by ASTP/ONC.

TEFCA and Digital Identity: TEFCA shows promise. We were in the first 10% of Epic customers to join the Epic Nexus Qualified Health Information Network (QHIN), and are live with three exchange purposes: treatment, public health and individual access service (IAS) with MyChart. However, broader participation and tighter standard operating procedures (SOPs) for individual access services identity proofing are needed. We are particularly interested in CMS's collaboration with other federal agencies and non-governmental partners to drive adoption and improve transparency and safety of digital identity solutions. We appreciate recent CMS announcements to address this concern for Medicare beneficiaries accessing Medicare services and systems and hope to see further expansion of such efforts. We understand that CMS will be modeling the use of digital identity at the national level and applaud its role in setting the standard.

Value-Based Care and Quality Reporting: We support CMS's move toward digital quality measures (dQMs) and FHIR-based reporting, particularly using Bulk FHIR. However, the reporting burden on health care providers is significant, and we have concerns about the use of certain electronic health records (EHR) data, which may be documented for billing purposes, as a true proxy for the quality of care delivered. We encourage CMS to consider the workflow burden on providers and to observe real-world clinical environments to better align reporting requirements with care delivery.

Patient Education and Engagement: Technology moves fast, and developing our nation's health literacy is a matter of both ensuring patients understand how to engage in their own health journey as well as how to take control of their health records. EHC established a Patient and

Family Advisor (PFA) program to ensure that our patients and their families receive compassionate care and that we are mindful of the needs and perspectives of patients and families as they navigate their care, including through interactions with health IT (e.g., patient portal, telehealth). Frequently, our informatics teams consult with the volunteer PFAs to ensure that projects align with patient and caregiver needs. We appreciate recent instances where CMS and its federal partners have sought out patient feedback (e.g., a recent CARIN Alliance Community Call that Dr. Thomas Keane, Amy Gleeson and Steven Posnack attended) and encourage CMS to continue seeking and supporting other ways patients can offer feedback and insights to all actors in the health IT ecosystem.

Building Trust Between Payors and Providers: Concerns about the patient's best interests are often part of discussions regarding the exchange of data from providers to patients (e.g., "Will this data be used to inappropriately deny or delay care to the patient?"). We encourage CMS to consider ways it can serve as a trust-builder between payors and providers. Without that trust, early adopters willing to test the application programming interfaces (APIs) included in CMS-0057 will likely encounter limitations.

REQUEST FOR INFORMATION: DETAILED RESPONSES

PC-12. What are the most valuable operational health data use cases for patients and caregivers that, if addressed, would create more efficient care navigation or eliminate barriers to competition among providers or both?

EHC supports patient-driven booking and management of appointments for certain services and believes this is a service that empowers patients in their care. We also believe patients are becoming increasingly familiar with quality rankings and metrics, especially when they are undergoing elective procedures. Transparency around metrics helps patients and their caregivers find high-quality care providers.

PC-10. How is TEFCA currently helping to advance patient access to health information in the real world?

At EHC, we support the Individual Access Services (IAS) Exchange Purpose (XP) with MyChart. We also have been pleased to support our nation's veterans by collaborating on the development of and committing our organization to the Veteran Interoperability Pledge. As an initial step of our participation, we have deployed the Veteran API to assist in accurately identifying veterans and are continuing to collaborate with the Department of Veterans Affairs (VA). We understand that the implementation of the government benefits determination XP may support similarly valuable interoperability at other organizations at scale.

Benefits like these require broader participation in TEFCA for meaningful patient access, as well as more requirements for digital identity verification. We recommend CMS consider:

- Tightening SOPs for identity proofing (see our response to PR-11).
- Encouraging vendor collaboration to improve transparency and safety.
- Supporting and, eventually, requiring additional TEFCA use cases.

- Investing in support for modernization and participation of public health systems in TEFCA exchange.

PR-2. What are obstacles that prevent development, deployment, or effective utilization of the most useful and innovative applications for physician workflows, such as quality measurement reporting, clinical documentation, and billing tasks? How could these obstacles be mitigated?
A lack of standardization in processes and documentation is a common barrier to effectively integrating data across multiple sources and the secondary use of that data (e.g., for research, public health, or quality measurement purposes). While process variation is natural amongst different providers, specialties, facilities and settings, there must be a path to integrating data in support of a holistic picture of the patient's health and wellbeing for their care providers. We expand on how data variation and poor-quality data sets impact the utility of that data in TD-13.

We also highlight in our answer to PR-5 the importance of support for all parameters of the Bulk FHIR IG in support of advancing population-level improvements in our nation's health and performant implementations. EHC is currently working to close care gaps and help patients manage their chronic conditions in collaboration with Guidehealth, a technology-driven health services platform that partners with health systems and clinical networks to bring scale and predictable performance to value-based care. Our collaboration with Guidehealth is focused on patients seen by affiliated providers the Emory Healthcare Network, our clinically integrated network with more than 3,940 physicians, including more than 475 primary care physicians. Bulk FHIR would be a potential technical solution for exchanging high volume data sets, including clinical notes, for this work. However, without support for all parameters of the Bulk FHIR IG and performant implementations, we must do additional work (e.g., creating multiple registries, running separate queries, investigating and relying upon vendor-specific standards) to achieve the scale of data exchange needed and expected through Bulk FHIR APIs.

We add that public health sources (e.g., Immunization Information Systems) are key sources of clinical data, and ensuring that they are FHIR-enabled and can be integrated with health care for the bidirectional exchange of information would support providers in closing care gaps and ensuring patients are up to date on preventative care services. On the topic of care gaps, we also encourage CMS to explore and clarify whether efforts to close care gaps or other population health activities for care coordination better align with "treatment" or "health care operations." The definitions today create limitations on our ability to use current data exchanges (e.g., TEFCA) to support caring for populations.

PR-5. Which of the following FHIR APIs and capabilities do you already support or utilize in your provider organization's systems, directly or through an intermediary? For each, describe the transaction model, use case, whether you use individual queries or bulk transactions, and any constraints:
As noted in our recent comments on CMS-1833-P, EHC has long supported health data interoperability, particularly through use of the Health Level Seven Fast Healthcare Interoperability Resources® (FHIR®) data standard. Creating a mature FHIR program has

required years of investment in process and workforce development. We share this history with you and others in the belief that it provides an example of a path to FHIR adoption across an organization.

To ensure that we established an operational structure that fully supported providing access to health data through FHIR, EHC established a FHIR Advisory Committee in July 2018. This 35-member committee was composed of representatives across Emory Healthcare and myriad disciplines, including physicians, advanced practice providers and nurses; IT, operations, patient access and marketing professionals; and members of the Patient Family Advisory Council. The FHIR Advisory Committee, established by the Chief Medical Information Officer and Chief Nursing Information Officer, served as the project steward for the organization's FHIR operations, strategy and governance program, working to establish project goals, measures of success and accountability, as well as removing barriers to project completion or schedule. With its inception, the committee promoted fiscal responsibility, efficiency of project completion, and authentic shared decision-making. At its core, its main goal was to advocate and advance the responsible and legally compliant access to and use of EHC data to support advancements in patient care, innovation and improvements in clinician workflow.

Over the course of five years, EHC commitment to FHIR-based health data interoperability took shape through the FHIR Advisory Committee, formal internal app development and for-license app evaluation and integration processes, processes for supporting patient requests to connect FHIR apps to their medical record, and collaborative, innovative health data interoperability-related activities with the Georgia Institute of Technology. With the FHIR program's maturity, EHC has now absorbed FHIR-related processes and operations into its broader digital governance and interoperability strategies.

Regarding our FHIR application programming interfaces (APIs): They have been in production for provider access since August 2018 and patient access since June 2019. We also support both EHR-launched and standalone app access. We do not currently have any active apps that use CDS Hooks but remain interested in starting to use them when appropriate.

As we also noted in our comments on CMS-1833-P, reducing data sharing burdens in health care, especially for Eligible Clinicians and Eligible Hospitals, requires capabilities for efficient bulk data exchange. This supports population-level health care operations as well as support for a growing population of patients who are quite sick or who have received care for many years and have large volumes of data in their medical records. This data is often essential for providing care and secondary, yet essential purposes. With tools that can exchange data in bulk, the potential for more efficient care management, faster research, simpler measurement of care quality and better understanding of opportunities to improve outcomes grows. Exchanges such as these are possible using the FHIR data standard (specifically, the Bulk FHIR specification), as evidenced by work performed at scale by leading health information exchanges.

However, many health care providers do not have access to Bulk FHIR APIs that perform to the necessary level for such benefits to be realized. EHC asks CMS to consider the following opportunities and challenges:

- Population level work requires that *all* parameters of the Bulk FHIR Implementation Guide be supported.
- Population level work also requires performant implementations of the APIs. This includes support for exports on true populations, not just cohorts of 1,000 patients. For example, EHC is participating in the Advanced Research Projects Agency for Health (ARPA-H) Biomedical Data Fabric, which leverages the Cumulus data pipeline to extract data from the EHR via Bulk FHIR. The solution gathers discrete data from the export and additional data via AI on clinical notes, and then deidentifies that the data and populates a dashboard for users with aggregate counts only. This is a tool with great potential for supporting population-level analysis and comparison of patient outcomes and research. However, we are having to segment small populations (4,600 patients) into 5 different registries to perform the Bulk FHIR-based data extraction.
- Standards must be in place and the utilization of those standards for syntactic and semantic interoperability tied to regulation (e.g., by requiring support for use of USCDI, with applicable vocabulary standards) can help ensure that organizations and developers do not default to the use of extensible resources, which hinders effective implementations. We have seen this create challenges in our own organization on projects leveraging Bulk FHIR and continue to invest resources in further mapping data elements beyond USCDI to standards, like LOINC or SNOMED.
- A standardized process for creating groups (i.e., not through vendor-specific methods, which is burdensome and not easily replicable) and performance requirements for large data pulls in a timely manner across populations would both support significantly greater utility of the Bulk FHIR APIs.

PR-8. What are ways CMS or partners can help with simplifying clinical quality data responsibilities of providers?

EHC is committed to providing the best care for our patients and works to improve quality through continuous pursuit of improvements in the quality of care provided by our providers. However, for certain quality measures, there is a real workflow burden of quality measurement on clinicians. Especially across systems, facilities or specialties where documentation workflows can vary significantly, standardized reporting can be challenging.

We encourage CMS to:

- Conduct site visits or virtual workflow studies to understand the impact of eCQMs and other quality measures.
- Fund additional research to evaluate existing quality measures, balanced with proposed measures, to ascertain the reliability of the measures (i.e., does the measurement accurately reflect quality outcomes and support movement towards improved quality of care) and the burden of documentation on both the patient and the provider.
- Fund explorations that might support organizations in preparing more comprehensive pictures of patient wellness and outcomes. As an organization, we are seeking to combine

all of our data sources into a Unified Data Platform, as we believe it will support our internal quality improvement and research efforts in the pursuit of the best care for our patients.

- Provide incentives or technical assistance for smaller systems to improve data quality as part of the journey to FHIR-based quality measures. Secondary uses of data, and data exchanges, rely on high-quality primary data sources.
- Additionally, we urge CMS to consider the unintended consequences of documentation burden on clinical care quality and provider burnout, and to incentivize innovative solutions that might reduce the introduction of technical friction between providers and patients.

PR-11. How could members of trust communities (for example, QHINs, participants and subparticipants in TEFCA, which requires Identity Assurance Level 2 (IAL2) via Credential Service Providers (CSPs)) better support the goals of reduced provider and patient burden while also enhancing identity management and security?

As we recently noted in our comments on CMS-1833-P, EHC strongly believes in the principles of a trusted exchange framework, provided there is diligent oversight and vigilance to ensure that participants utilize the exchange purposes appropriately. EHC believes being a responsible participant in TEFCA requires actors involved in health data exchange to establish strong operational ownership and maintain ongoing processes. EHC also holds that responsible participation in TEFCA also requires that technical standards or standard operating procedures (SOPs) support assurances of trust and verification. EHC thus encourages ASTP/ONC to revise the Individual Access Services (IAS) XP to add additional demographic data points (e.g., sex, email address, and last four digits of an individual's Social Security Number) to the required data elements cited in 4.5.1. Verification Demographics, section A, of the TEFCA IAS XP SOP.

TEFCA is not a one-time initiative, but rather an ongoing commitment. The true value of TEFCA will be realized through broader participation, and EHC encourages CMS, alongside other HHS agencies, to continue its efforts in promoting TEFCA with a long-term vision for its success. The value to patients will not be realized until there is a sufficient participant base amongst providers and other health care actors.

EHC also encourages CMS to explore ways to collaborate with developers of health IT (specifically, EHR technologies) to better communicate the safety and trustworthiness of data exchanges that follow federally endorsed digital identification methods.

TD-7. To what degree has USCDI improved interoperability and exchange and what are its limitations?

USCDI has improved baseline interoperability, and EHC supports the continued development of USCDI and USCDI+ data sets to further expand the floor of nationwide interoperability. We were particularly pleased to see the recent award announcement for the piloting of USCDI+ for behavioral health and thank CMS for the incentivization for historically underfunded health care providers to modernize their systems and improve their native data quality using common data models.

Certain limitations do exist in real-world use of USCDI and the standards it maps to. For example, ICD and CPT codes can be inaccurate, which creates downstream challenges when repurposed for research or public health. Our work to improve the flow of data between EHC and the Georgia Emerging Infections Program (EIP) via FHIR has shown that ICD-10 codes alone are insufficient for case identification; surveillance epidemiologists often rely on clinical notes instead or in addition to coded data elements, but review of those notes requires significant manual work to extract insights in the absence of large language model (LLM)-enabled rapid analysis. We are leveraging findings from our initial implementation of a FHIR application for the exchange of this information to seek additional funding that would introduce artificial intelligence/machine learning (AI/ML) into these workflows and working to address any mapping gaps in our own data sets.

We recommend:

- Continued investment in the USCDI and USCDI+ models, especially for smaller health care organizations.
- Consideration of incentives or requirements for mapping data to and beyond the floor of interoperability for true data exchange capabilities and improved data quality.
- Support of tools to assess data quality at the element level.

TD-13. What new opportunities and advancements could emerge with APIs providing access to the entirety of a patient's electronic health information (EHI)?

Full EHI access enables continuity of care, reduces duplicate testing, and supports real-time decision support. For example, patients moving between states or transitioning from pediatric to adult care often face delays due to inaccessible historical records. A truly interoperable EHI ecosystem would allow providers to access prior diagnoses, labs, and imaging—improving outcomes and reducing costs.

However, this requires:

- Data quality certification programs.
- Further expansion and adoption of common data definitions and terminologies (e.g., timestamp standardization, provenance).

At EHC, we are moving toward a Unified Data Platform (UDP). This includes incorporating EHI data from EHR, imaging, devices (e.g., real-time monitors), financial systems, and more. Developing a platform with all this information in a single ecosystem, with accurate linkages between the information, is a resource-intensive process. Throughout this process, we have discovered areas that would be particularly beneficial to have support from federal agencies in standardization. For example, our work with ICU waveform data and real-time device integration underscores the need for standardization and better support for high-volume, high-frequency data types, perhaps in partnership between CMS, ASTP/ONC and FDA.

The transition to a UDP at our organization is greatly enhanced by the use of AI/ML, but clean, timely data is required. This requires that we all, as an ecosystem, "walk before we run." This is detail-oriented work.

For AI to be effective:

- Data must be AI-ready (clean, structured, timely). Standards are a key component of delivering high-quality data to all end users.
- Guardrails and federal guidelines must be developed and enforced in the governance of AI/ML in the context of health data and health care.

TD-18. Information blocking: What additional policies could ASTP/ONC and CMS implement to further discourage healthcare providers from engaging in information blocking practices?

We support maintaining and strengthening the Information Blocking Rule. We encourage CMS to continue promoting standards (including standardized APIs) that support vendor-agnostic data sharing.